MIL-W-13871B
14 July 1966
SUPERSEDING
MIL-W-13871A(Ord)
18 April 1961
(See 6.3)

MILITARY SPECIFICATION

WOOD BLANKS FOR SMALL ARMS

This Specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers wood blanks for gunstocks, buttstocks, handguards, and similar small arms parts.

1.2 Classification.

1.2.1 Types. Wood blanks shall be furnished in the following types, as specified (see 3.1 and 6.1):

Type I - Walnut
Type II - Birch

1.2.2 Condition. Wood blanks shall be furnished in the following conditions, as specified (see 6.1):

Kiln-dried
Air-dried
Green

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-388 - Plastic Sheet and Strip (Polyolefin)

FSC 5510

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STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Inspection Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized and Containerized Unit Loads

HANDBOOKS

MIL-HDBK-7 - Military Handbook, Lumber and Allied Products

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Species.

3.1.1 Type I. Walnut shall be either black walnut (Juglans nigra) or English walnut (Juglans regia). When English walnut is furnished, it shall have a minimum specific gravity of 0.51 at 6 to 8 percent moisture content, when tested as specified in 4.6.2.1.

3.1.2 Type II. Birch shall be either yellow birch (Betula alleghaniensis) or sweet birch (Betula lenta).

3.2 General characteristics. Blanks shall be free of unsound or brashy wood, checks, splits, shakes, honeycombing, and excess warpage or shrinkage. The following defects will be permitted: sapwood, mineral streaks, figured or burly grain, scattered sound pin knots not exceeding three-eights inch average diameter and not more than three per blank, and scattered pinworm holes not exceeding one-sixteenth inch diameter and not more than six per blank. Figured or burly grain, knots, and pinworm holes shall not be permitted in the critical area as shown in Figure 1. Defects in excess of the limitations specified above will be permitted provided their location (in other than the critical area) is such that they will be removed during the fabrication of the finished product.

3.3 Grain. The slope of the grain shall be in accordance with Figure 1. When both spiral and diagonal slope of grain exist, the combined slope of grain shall not be greater than 1 in 12.

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3.4 Moisture content.

3.4.1 Kiln-dried blanks. The average moisture content of kiln-dried blanks shall not be less than 6 percent nor more than 8 percent when tested as specified in 4.6.2.2. The gradient of moisture content between core and outer shell shall not vary more than 2 percent.

3.4.2 Air-dried blanks. Air-dried blanks shall have an average moisture content of not more than 25 percent when tested as specified in 4.6.2.2.

3.4.3 Green blanks. There is no specified minimum or maximum moisture content for green blanks and testing therefore is not required.

3.5 Casehardening. When blanks are furnished in the kiln-dried condition, test specimens tested as specified in 4.6.2.3 shall be free of severe casehardening as illustrated in Figure 2; slight casehardening (i.e., casehardening to the extent that the prongs of the test specimen do not touch each other) shall be permitted in no more than two of the test specimens.

3.6 End coating. Both ends of each blank, including test samples, shall be thoroughly coated immediately after being cut to final blank form. Any commercial type end coating which completely seals end grain and effectively controls end splitting shall be used except that coatings containing varnish or powdered metal, such as powdered aluminum shall not be permitted. Coatings shall not extend onto the sides of the blank more than three-quarters of an inch from either end.

3.7 Dimensions. Blanks shall conform to the dimensions and tolerances specified in the contract or purchase order (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection terms and definitions. Inspection terms and definitions used herein are in accordance with MIL-STD-109.

4.2 Classification of inspection. All examination and testing in this specification shall be to determine conformance to the requirements of the specification and to serve as a basis for acceptance of the material covered by this specification.

4.3 Lot. A lot of kiln-dried blanks shall consist of all pieces of the same size and type from one kiln load. For material not kiln dried, a lot shall consist of all blanks of the same size, type and condition presented for acceptance at one time.

4.4 Sampling.

4.4.1 For visual and dimensional examination. Sample size shall be in accordance with MIL-STD-105, using Inspection Level I.

4.4.2 For specific gravity determination of English walnut. Ten blanks shall be taken from each lot and tested for specific gravity using the test methods specified in 4.6.2.1.

4.4.3 For moisture content and casehardening testing of kiln-dried blanks. Fifty-four full size blanks shall be used as kiln samples for each kiln charge. Eighteen test blanks shall be identified for test, and 36 for retest, if required. The samples shall be placed in groups of three in each kiln charge. Each group shall contain one blank for test and 2 blanks for retest. The groups shall be located to represent blanks in the fastest drying and slowest drying areas in the charge and shall be inserted in such a manner that they will be subjected to the same drying conditions as the blanks surrounding them. After completion of kiln drying, the samples identified for test shall be tested for moisture content and casehardening using the applicable test methods specified in 4.6.2.2 and 4.6.2.3 respectively.

4.4.4 For moisture content testing of air-dried blanks. Ten blanks shall be taken from each lot and tested for moisture content using the applicable test method specified in 4.6.2.2. Samples shall be selected to include what appears to be the wettest material in the lot.

4.5 Examination.

4.5.1 Visual and dimensional. Examination of blanks shall be performed in accordance with 4.5.1.1 on each of the sample blanks taken in accordance with 4.4.1.

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4.5.1.1 Classification of defects. Classification of defects and acceptable quality levels (AQL's) for examination shall be as specified herein and shall include the following provisions:

(a) The AQL's are specified as percent defective.
(b) The AQL specified for each defect number shall be applied to the defect or group of defects listed for that defect number.

Categories	Defects	AQL	Inspection Method
Critical	None defined		
Major			
101	Unsound or brashy wood, figured or burly grain, checks, splits, shakes, knots, or pinworm holes (see 3.2)	0.65	Visual
102	Improper contour or wave (see 3.7)	1.0	Gage
103	Improper slope of grain (see 3.3)	1.0	Visual
Minor			
201	Missing or improper end coating (see 3.6)	1.5	Visual
202	Thickness, under minimum (see 3.7)	1.5	Measure
203	Excessive warpage or shrinkage (Sample blanks which are visually detectable as warped or shrunk shall be classified as defective. When the number of defective blanks exceeds the rejection number, the contractor may demonstrate usuability of the defective blanks by screening and correcting the entire lot from which the sample has been taken. The blanks may be planed on both sides to not less than the minimum thickness specified in the contract or purchase order. Blanks which have been planed and are acceptable may be returned to the		

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Categories	(Continued) Defects	AQL	Inspection Method
	represented lot. If in a sampling of the screened and corrected lot, the number of defective blanks exceeds the rejection number, the lot shall be rejected. If the number of defective blanks is less than the rejection number the lot may be accepted.)	1.5	Visual
204	Checks, splits, shakes or honeycombing on ends of blanks (After all other examinations have been performed, approximately 1/8 to 1/4 inch shall be cut off each sample blank and the blank ends visually examined for checks, splits, shakes or honeycoming. When the number of defective blanks exceeds the rejection number, the entire lot will be returned to the contractor who shall cut off 1/8 to 1/4 inch from each end of each blank, screen the entire lot, recoat the ends of acceptable blanks, and return them to the lot from which they were taken.)	2.5	Visual
205	Thickness, over maximum (see 3.7)	4.0	Measure

4.5.2 Preparation for shipment. Examination shall be made to determine compliance with the requirements of Section 5.

4.6 Testing.

4.6.1 Test specimens.

4.6.1.1 Specific gravity of English walnut.

4.6.1.1.1 Kiln-dried blanks. One specimen approximately 2 by 2 inches shall be taken from each sample blank (see 4.4.2).

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4.6.1.1.2 Air-dried or green blanks. Each sample blank taken in accordance with 4.4.2 shall be dried to 6 to 8 percent moisture content. After drying, one specimen approximately 2 by 2 by 6 inches shall be taken from each blank. The moisture content of the dried sample blanks shall be determined in accordance with 4.6.2.2 and 4.6.2.2.2.

4.6.1.2 Moisture content.

4.6.1.2.1 Kiln-dried blanks. One specimen shall be taken from each of the 18 test blanks (see 4.4.3) in accordance with Figure 3. One-quarter of the length of each specimen shall be cut from each end and discarded, and the remaining middle portion shall be cut so as to remove one-quarter of the original width from each side, as illustrated in Figure 3. The two side pieces shall be used in determining the moisture content of the shell and the center portion in determining the moisture content of the core.

4.6.1.2.2 Air-dried blanks. Specimens shall be taken from each sample blank (see 4.4.4) in accordance with Figure 4. The whole test specimen shall be used for determination of moisture content.

4.6.1.3 Casehardening of kiln-dried blanks. Specimens shall be taken from each of the 18 test blanks (see 4.4.3) and prepared in accordance with Figure 3.

4.6.2 Test methods.

4.6.2.1 Specific gravity. Test specimens prepared in accordance with 4.6.1.1 shall be weighed and measured immediately after cutting. The weights shall be determined to an accuracy within ±0.2 percent and the measurements to an accuracy ±0.3 percent. The specific gravity shall be calculated as follows:

$$\text{Specific gravity} = \frac{K(\text{or } K_1)W}{LWT}$$

Where:

W = Weight of specimen at 6 to 8 percent moisture content, in grams.
L = Length of specimen, in inches or centimeters.
T = Thickness of specimen, in inches or centimeters.
K = 1, when metric units of weight and measurement are used.
K_1 = 0.061, to be used only when metric units of weight and English units of measurement are used.

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4.6.2.2 Moisture content. Moisture content shall be determined by the oven-drying method. Electric moisture meters will not be permitted unless specifically authorized by the contracting officer. Test specimens shall be weighed immediately after cutting, then dried in an oven at 103° ± 2° centigrade until approximately constant weight is attained. After drying, specimens shall be weighed as soon as they attain room temperature. The weights shall be determined to an accuracy within ± 0.2 percent. The moisture content shall be calculated as follows:

$$\text{Moisture content (percent)} = \frac{W-F}{F} \times 100$$

Where:
W = Initial weight of specimen
F = Final weight of specimen

4.6.2.2.1 Kiln-dried blanks. Determinations shall be made using the shell and core sections prepared as specified in 4.6.1.2.1. The moisture content of the two side pieces together shall be determined and recorded as that of the shell. The moisture content of the center portion shall be determined separately and recorded as that of the core. The average of the shell reading and the core reading shall be recorded as that of the blank.

4.6.2.2.2 Air-dried blanks. The moisture content of the whole test specimen taken in accordance with 4.6.1.2.2 shall be determined and recorded as that of the blank.

4.6.2.3 Casehardening. Test specimens prepared in accordance with 4.6.1.3 shall be allowed to come to a uniform moisture condition of not more than 10 percent and not less than 6 percent and visually examined for evidence of casehardening (see 3.5 and Figure 2).

4.7 Rejection.

4.7.1 Examination. Unless otherwise specified herein, individual blanks not meeting the requirements of this specification shall be rejected. The represented lot shall be rejected when the number of rejected sample blanks equals or exceeds the rejection number specified in MIL-STD-105 for the sample size of the lot under examination.

4.7.2 Testing. Failure to comply with any of the test requirements of this specification shall be cause for rejection of the represented material.

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5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level C. Blanks shall be packed in a manner to insure carrier acceptance and safe delivery at destination.

5.1.1.1 Air-dried or green wood blanks shall be delivered in self contained units as shown in Figure 5.

5.1.1.2 Kiln-dried blanks shall be delivered in self contained units as shown in Figure 5. In addition, kiln-dried blanks shall be protected from the weather by being warpped as described in MIL-STD-147 (Storage Aid No. 9) with the exception that polyethylene film meeting Federal Specification L-P-378, type II, class 1, grade B, finish 1 shall replace the duplex asphaltic laminated waterproof paper (UU-P-271) specified. Rail or truck shipments shall be made in box cars or closed trucks.

5.2 Marking. In addition to any special marking required by the contract or order, self contained unit loads shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2.1).
(c) Condition required (see 1.2.2).
(d) Size, dimensions and tolerances required (see 3.7).

6.2 Definitions. Definitions of terms relating to wood characteristics and defects used in this specification may be found in Military Handbook 7 "Lumber and Allied Products".

6.3 Supersession data. This specification also supersedes Springfield Armory Purchase Description SAPD-200.

Custodians:
Army - WC
Navy - OS
Air Force - None

Preparing activity:
Army - WC

(Project Nr 5510-0098)

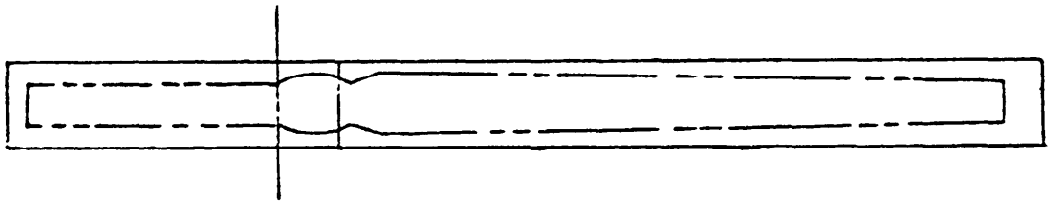
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Review activities:
Army - None
Navy - None
Air Force - None
DSA - CS

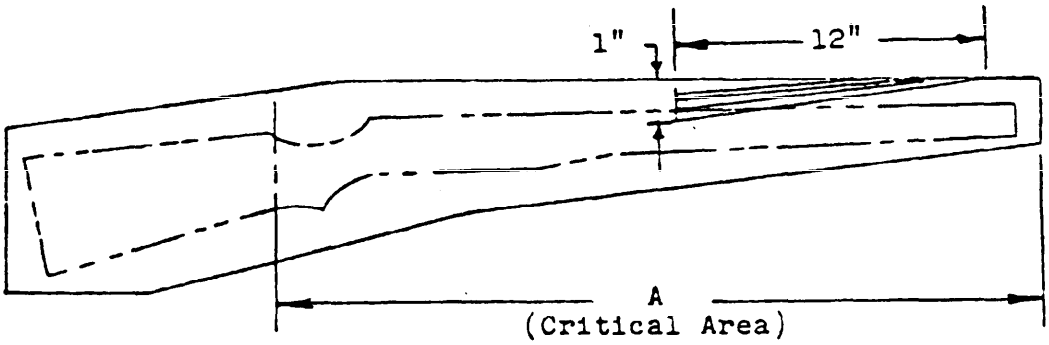
User activities:
Army - None
Navy - MC
Air Force - None

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Grain shall be defined as the direction, arrangement, and appearance of the fibers which form the substance of the wood.



Top View. Slope of grain in the critical area shall not exceed 1 inch in 12 inches measured parallel with either side of blank.

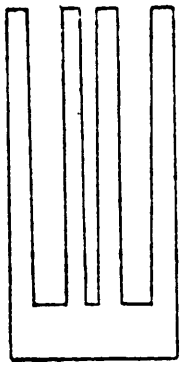


Side View. Grain direction in the critical area may vary from horizontal to a slope of 1 inch in 12 inches in direction shown.

Stock Blank Dimension	A
Rifles: M1, M1C, M1D, & M14	26 1/2" ± 1"
Carbines: M1, M2, & M3	22" ± 1"

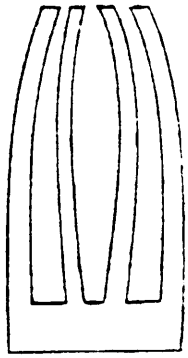
Figure 1. Grain Slope in Critical Area of Stock Blanks

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Not

Casehardened

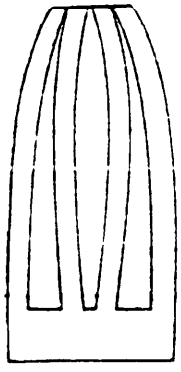


Slightly

Casehardened

Acceptable

(see 3.5.)



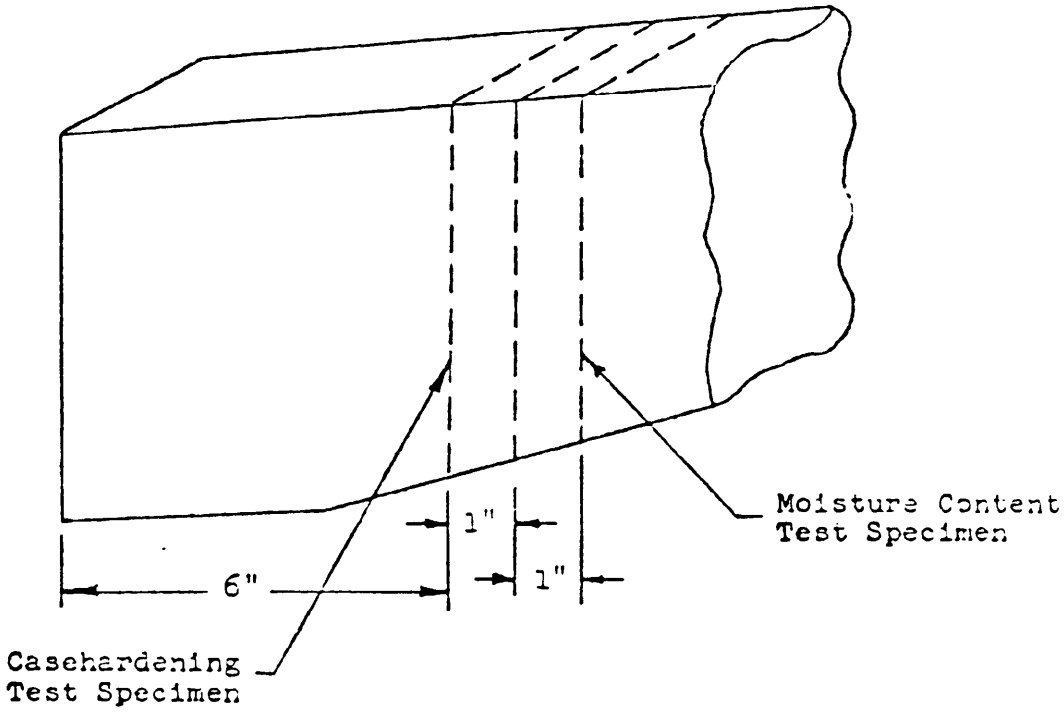
Severely

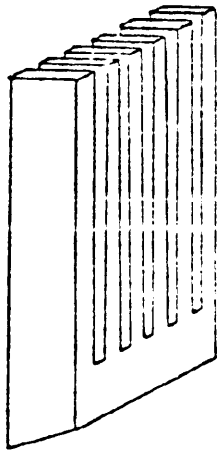
Casehardened

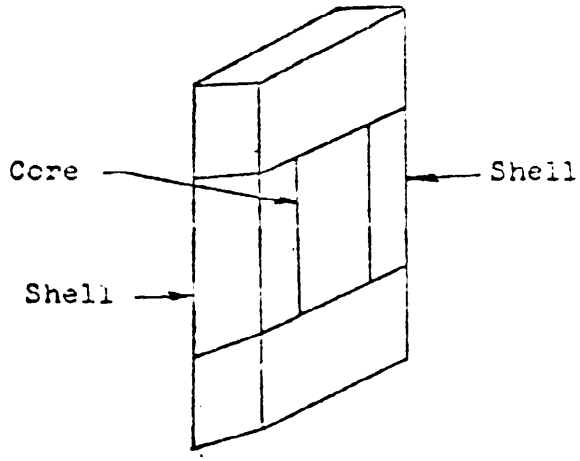
Not Acceptable

Figure 2. Casehardening Test Standards

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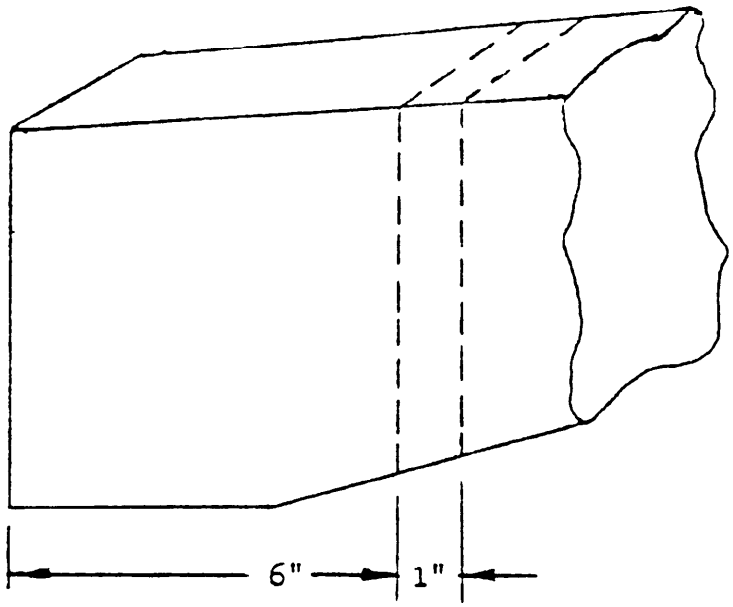




Specimen shall be sawed as shown so as to produce six prongs of equal thickness: Prongs 2 and 5 shall be broken out.

Figure 3. Casehardening and Moisture Content Test Specimens from Kiln-Dried Blank

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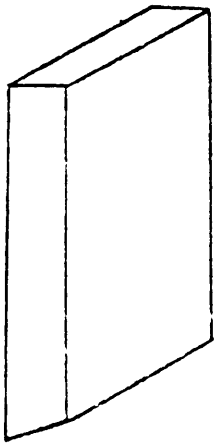


Figure 4. Moisture Content Test Specimen from Air-Dried Blank

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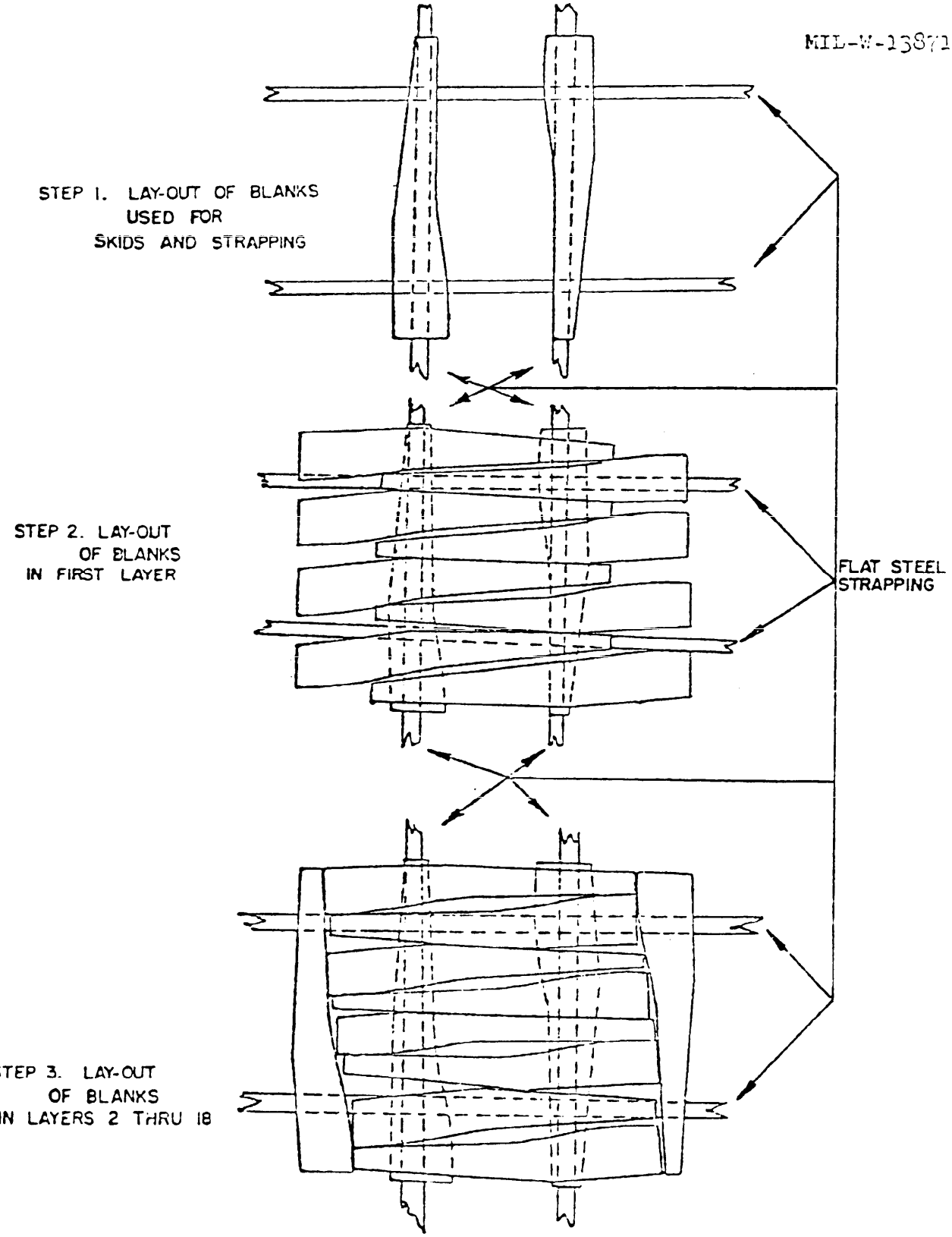


FIGURE 5. GUNSTOCK BLANK PACKAGE

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (*Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity*)

SUBMITTED BY (*Printed or typed name and activity*)	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE

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